REPORT
Customer Churn Analysis and Retention Strategy

1. Project Overview

This project provides a comprehensive overview of customer churn analysis in a telecom company using Python, SQL, and Power BI. Data preprocessing and exploratory analysis were performed in Python, while SQL queries evaluated churn across customer demographics, tenure, contract types, and service usage. An interactive dashboard was developed to visualize key patterns and compare churned and retained customers. **The analysis identified major factors influencing churn, and the findings were used to suggest data-driven improvements to reduce churn and enhance customer retention based on observed customer behavior.**

2.Dataset Summary

- Total Rows: 7043 Total Columns: 21
- Dataset Type: Customer Churn Analysis Dataset
- Numerical Columns: Senior Citizen, tenure, MonthlyCharges, TotalCharges , Rating

3.Exploratory Data Analysis Using Python

Data Preparation and Cleaning
We began with data preparation and cleaning using Python.

Data Loading: Imported the dataset using the pandas library.
Initial Exploration: Used df.info() to understand the structure of the dataset and df.describe() to generate summary statistics for numerical features.

```
dp.head()
```

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	...
0	7590-VHVEG	Female	0	Yes	No	1	No	No phone service	DSL	No	...
1	5575-GNVDE	Male	0	No	No	34	Yes	No	DSL	Yes	...
2	3668-QPYBK	Male	0	No	No	2	Yes	No	DSL	Yes	...
3	7795-CFOCW	Male	0	No	No	45	No	No phone service	DSL	Yes	...
4	9237-HQITU	Female	0	No	No	2	Yes	No	Fiber optic	No	...

5 rows × 21 columns

```
dp.describe()
```

	SeniorCitizen	tenure	MonthlyCharges	TotalCharges
count	7043.000000	7043.000000	7043.000000	7043.000000
mean	0.162147	32.371149	64.761692	7274.286334
std	0.368612	24.559481	30.090047	2423.045693
min	0.000000	0.000000	18.250000	0.000000
25%	0.000000	9.000000	35.500000	8196.400000
50%	0.000000	29.000000	70.350000	8196.400000
75%	0.000000	55.000000	89.850000	8196.400000
max	1.000000	72.000000	118.750000	8468.200000

```
dp.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   customerID        7043 non-null   object
 1   gender            7043 non-null   object
 2   SeniorCitizen     7043 non-null   int64
 3   Partner           7043 non-null   object
 4   Dependents        7043 non-null   object
 5   tenure            7043 non-null   int64
 6   PhoneService      7028 non-null   object
 7   MultipleLines     7043 non-null   object
 8   InternetService   7043 non-null   object
 9   OnlineSecurity    7043 non-null   object
 10  OnlineBackup      7043 non-null   object
 11  DeviceProtection  7043 non-null   object
 12  TechSupport       7043 non-null   object
 13  StreamingTV       7043 non-null   object
 14  StreamingMovies   7043 non-null   object
 15  Contract          7043 non-null   object
 16  PaperlessBilling  6928 non-null   object
 17  PaymentMethod     7043 non-null   object
 18  MonthlyCharges    7043 non-null   float64
 19  TotalCharges      7043 non-null   float64
 20  Churn             7043 non-null   object
dtypes: float64(2), int64(2), object(17)
memory usage: 1.1+ MB
```

- **Missing Data Handling and Duplicate Values:**Checked for null values and duplicate values

```
print(dp.isnull().sum())

customerID            0
gender                0
SeniorCitizen         0
Partner               0
Dependents            0
tenure                0
PhoneService         15
MultipleLines         0
InternetService       0
OnlineSecurity        0
OnlineBackup          0
DeviceProtection      0
TechSupport           0
StreamingTV           0
StreamingMovies       0
Contract              0
PaperlessBilling    115
PaymentMethod         0
MonthlyCharges        0
TotalCharges          0
Churn                 0
dtype: int64
```

```
dp[['PhoneService','PaperlessBilling']].mode()

   PhoneService  PaperlessBilling
0           Yes               Yes

update_column=dp
up_val=['PhoneService','PaperlessBilling']
update_column[up_val]=update_column[up_val].fillna(dp[['PhoneService','PaperlessBilling']].mode().iloc[0])
```

```
dp.duplicated().sum()

np.int64(0)
```

4.Data Analysis using SQL

Performed structured analysis in PostgreSQL to answer key Problem Questions

1.How many senior citizens vs non-senior

	seniorcitizen integer	count bigint
1	0	5901
2	1	1142

2.Customers by Gender

	gender character varying (10)	count bigint
1	Female	3488
2	Male	3555

3.How many customer churned or stayed

	churn character varying (5)	yes_or_no bigint
1	No	5174
2	Yes	1869

4.Which contract type has the highest churning rate

	contract character varying (25)	total_contract bigint	churned_case bigint	churn_rate numeric
1	One year	1473	166	11.27
2	Month-to-month	3875	1655	42.71
3	Two year	1695	48	2.83

5.Average monthly Charges vs Churn

	churn character varying (5)	avg_monthly_charges numeric
1	No	61.2651236954000773
2	Yes	74.4413322632423756

6.Churn Rate by Monthly charges Range

	charge_range text	total_customers bigint	churned_customers bigint	churned_rate numeric
1	0-30	1653	162	9.80
2	30-50	641	199	31.05
3	50-70	1158	234	20.21
4	70-90	1847	701	37.95
5	90+	1744	573	32.86

7.How does customer churn vary across different internet service types?

	internetservice character varying (20)	total_customers bigint	churned_customers bigint	churn_rate numeric
1	No	1526	113	7.40
2	DSL	2421	459	18.96
3	Fiber optic	3096	1297	41.89

8.How does churn vary for customers with both online security and device protection

	onlinesecurity character varying (25)	deviceprotection character varying (25)	total_cutomer bigint	churned bigint	churn_rate numeric
1	No	No	2187	1029	47.05
2	No	Yes	1311	432	32.95
3	Yes	No	908	182	20.04
4	Yes	Yes	1111	113	10.17
5	No internet service	No internet service	1526	113	7.40

9.How Tech support helps Customers retention

	techsupport character varying (25)	total_support bigint	total_churned bigint	churn_rate numeric
1	No	3473	1446	41.64
2	No internet service	1526	113	7.40
3	Yes	2044	310	15.17

10.Paperless billing impact on churning rate

	paperlessbilling character varying (5)	billing bigint	total_churned bigint	churn_rate numeric
1	No	2828	460	16.27
2	Yes	4215	1409	33.43

11.Dependents vs Churn

	dependents character varying (5)	total_dependents bigint	total_churned bigint
1	No	4933	1543
2	Yes	2110	326

12.Impact of payment method on churning rate

	paymentmethod character varying (50)	total bigint	churned bigint	churn_rate numeric
1	Electronic check	2365	1071	45.29
2	Mailed check	1612	308	19.11
3	Bank transfer (automat...	1544	258	16.71
4	Credit card (automatic)	1522	232	15.24

KEY INSIGHTS:-

1.Contract Type Influences Churn
Customers with month-to-month contracts show the highest churn compared to long-term contract customers.

2.Monthly Charges Impact Customer Retention
Higher monthly charges are associated with increased churn rates.

3.Tenure Plays a Major Role
Customers with shorter tenure are more likely to churn, while long-term customers remain loyal.

4.Support Services Reduce Churn
Customers without Tech Support, Online Security, or Device Protection have higher churn.

5.Payment Method Affects Churn Behavior
Customers using electronic check tend to churn more than those using automatic payment methods

SUGGESTIONS

1.Encourage Long-Term Contracts
Promote yearly and two-year plans to reduce churn among month-to-month customers

2.Optimize Pricing for High-Charge Customers
Review high monthly charges and provide value-added benefits to retain customers.

3.Focus on New Customers
Improve early customer experience since churn is higher during initial tenure.

4.Promote Support Services
Increase awareness of Tech Support, Online Security, and Device Protection services

5.Improve Payment Convenience
Encourage customers to use automatic payment methods for better retention.